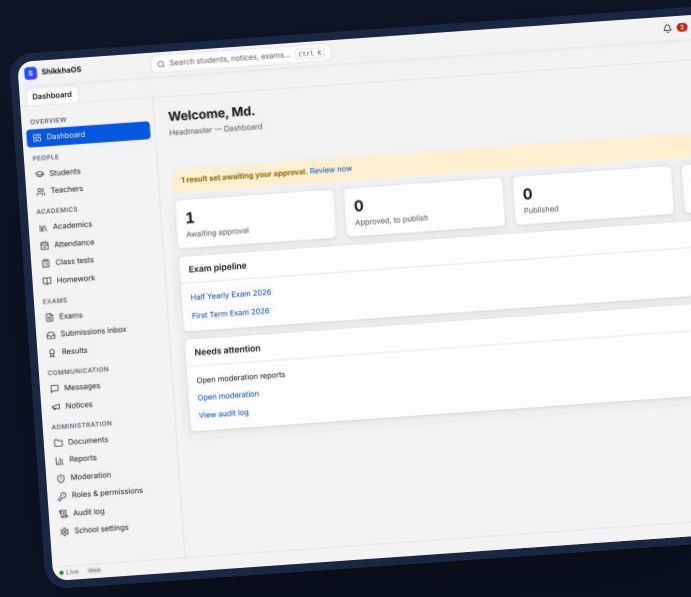
ShikkhaOS
SCHOOL OPERATING SYSTEM

One school. One system.

A single, secure digital platform for administration, teaching, examinations and communication — built for our school.



Project Proposal
For the School Managing Committee

November 2026
Version 1.0 — for decision

Platforms
Windows desktop · Web · Android

Web petrait.site Phone 01616154320 Email petraitsol@gmail.com

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What is inside

Read pages 3–7 for the idea, pages 8–13 to see the system working, and pages 14–19 for safety, status and the decisions we ask for.

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> **“One School. One System. One Connected Community.”**

EXECUTIVE SUMMARY

What we are proposing

ShikkhaOS replaces scattered registers, spreadsheets and chat groups with one connected school system that every teacher, the Register, the Headmaster and every student can use — each seeing only what they should.

3	4	7	1
apps: desktop, web, Android	user roles with separate powers	safe steps from marks to published result	permanent record of every change

What the school gets

- ✔ Marks entered once, checked by the Register, approved by the Headmaster, then published to students.
- ✔ Daily attendance, class tests and homework in the same place.
- ✔ Notices and messages that reach the right people — with student-safety rules built in.
- ✔ Reports, report cards and certificates generated from the same data.

What the committee decides

This document shows the real system working, explains how it protects students and results, and asks for six decisions (last pages):

1. Approve a **pilot** in one class.
2. Choose **where the system runs**.
3. Confirm **who may publish results**.
4. Nominate a **system owner** and training time.
5. Confirm the **messaging policy**.
6. Confirm **data handling** and backups.

About the pictures. Every screen in this document is a real screenshot of the working system. The names, marks and notices shown are invented demonstration data — no real student information is used.

WHY CHANGE

Common problems — and what changes

These are problems schools typically face when work lives on paper and in chat groups. We would confirm which apply here during the pilot.

Typical problem today	With ShikkhaOS
Marks are written on sheets, copied to registers and re-typed for the result sheet; small errors slip through.	Teachers enter marks once. The system checks the range (0 to full marks), totals them and calculates grades and positions.
Nobody can tell who changed a mark, or when.	Every change is recorded with who, what and when. Submitted marks are locked; corrections go through a visible request.
Results are delayed while sheets travel between teachers, Register and Headmaster.	Each person sees exactly what is waiting for them. The Headmaster sees pending approvals on one screen.
Students learn results late or by word of mouth.	When the Headmaster publishes, students are notified and see their own result and report card — no one else's.
Attendance registers are checked rarely; absences are noticed late.	Teachers take attendance in seconds. Attendance percentages are always up to date for each student and class.
Notices and questions go through informal chat groups with no school control.	Official notices and school-managed messaging. Private chat between students is not available; reports and moderation exist.
Staff access to records is informal.	Role-based access: a teacher sees only their own classes; the Register and Headmaster have different powers.

Built for one school

ShikkhaOS is a private system for our school only — not a shared product. Our data stays under the school's control.

No parent accounts (first release)

Students use the app. Guardian contact details are stored as records only; parents do not log in during this first release.

THE SYSTEM

One school server, three ways in

Everyone uses the same data through the app that suits them. Nothing is typed twice.

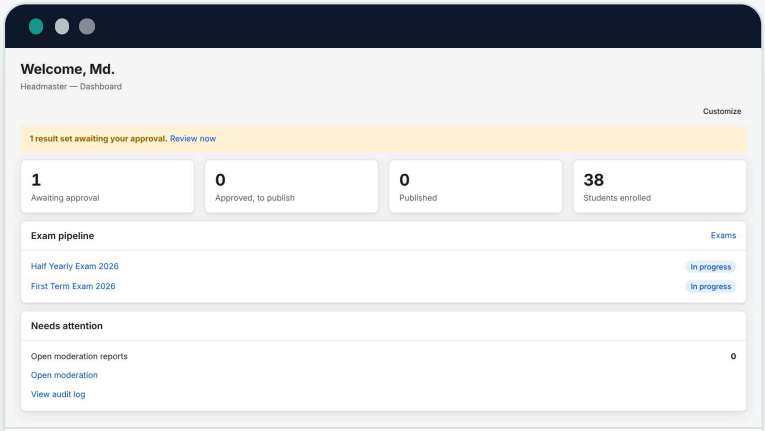
Windows desktop app
Headmaster · Register · Teachers

Web app (any browser)
All roles

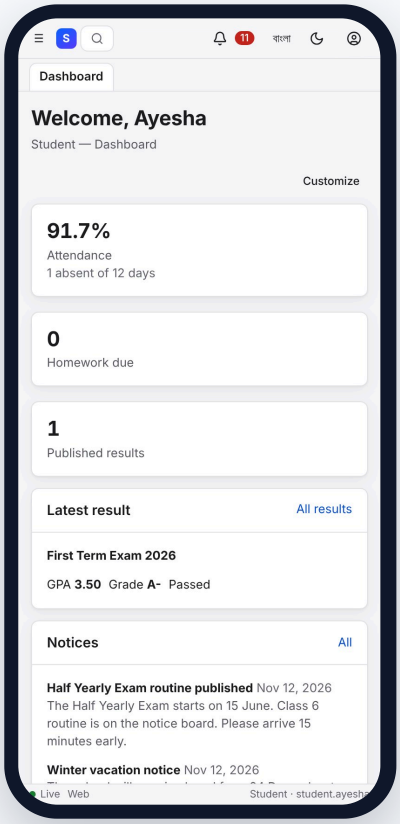
Android app
Students

⇄

ShikkhaOS school server
Rules · permissions · database · audit record



Headmaster dashboard (desktop/web): what needs attention, at a glance.



Student view on a phone

Desktop

A proper Windows application for staff: quick search (Ctrl+K), tabs, notifications, light/dark theme.

Web

Works in any modern browser with the same screens and rules — handy on a shared office computer.

Android

A student app for results, attendance, homework, notices and messages, with notifications.

PEOPLE AND POWERS

Who does what

Four kinds of users. What each person can do is set by the school, and enforced by the server — hiding a button is never the only protection.

Headmaster

- School-wide dashboard and reports
- Authorizes exams; **approves and publishes** results
- Sets who may do what (permissions)
- Official announcements, moderation, audit log

Register

- Admissions, student and teacher records
- Sets up exams; **checks marks** from teachers
- Requests corrections; compiles results
- Certificates, report cards, documents
- Publishes only if the Headmaster delegates it

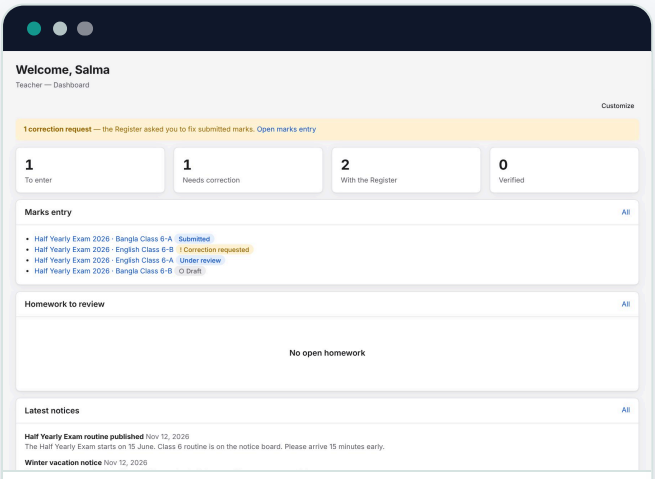
Teacher

- Sees **only their own** classes and subjects
- Attendance, marks entry, class tests, homework
- Submits marks to the Register; fixes corrections
- Messages colleagues, own class group

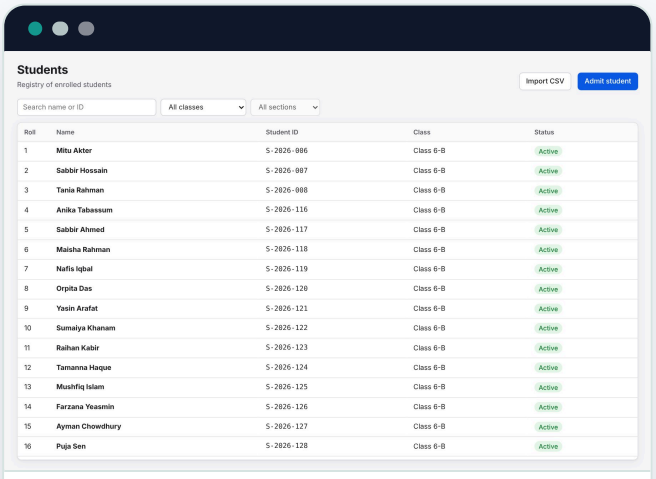
Student

- Sees **only their own** record
- Published results and report cards
- Attendance, class tests, homework, notices
- Messages to their own teachers (school-controlled)

Headmaster and Register are deliberately different. The Register prepares and verifies; the Headmaster approves. The person who verified marks cannot also approve them, and approving or publishing needs the password to be typed again.



Teacher dashboard: a correction request from the Register is highlighted.



Register: student registry with search and class filters.

A DAY IN THE LIFE

One school day, seen by each person

An illustrative day, using the demonstration school. It shows how the pieces fit together without anyone retyping anything.

8:05 · TEACHER

Takes attendance in seconds

Roll call on a tablet or computer; absences are counted straight away.

10:30 · TEACHER

Gives homework

Posted to the class; students see it in the app.

12:00 · TEACHER

Submits exam marks

Range-checked, totalled; locked once submitted.

13:15 · REGISTER

Checks the submission

Spots a doubtful total and requests a correction with a reason.

14:00 · TEACHER

Fixes and resubmits

Saved as a new revision; the old one is kept.

15:30 · HEADMASTER

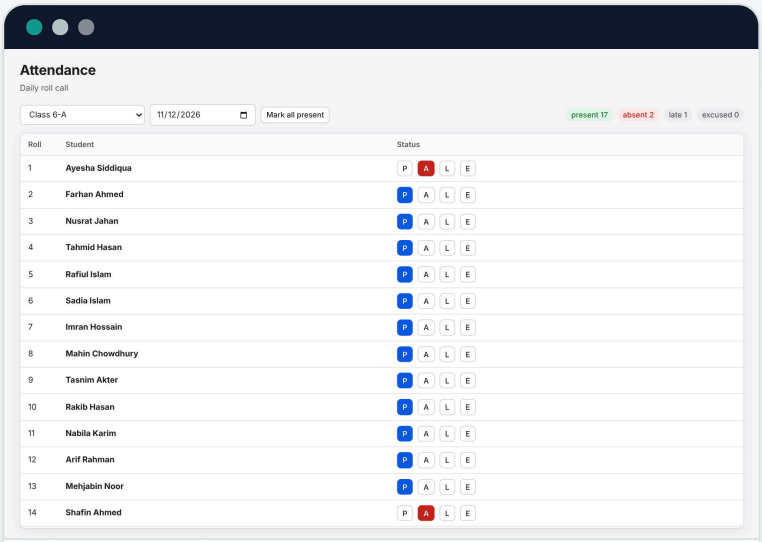
Approves and publishes

Password typed again; the decision is recorded.

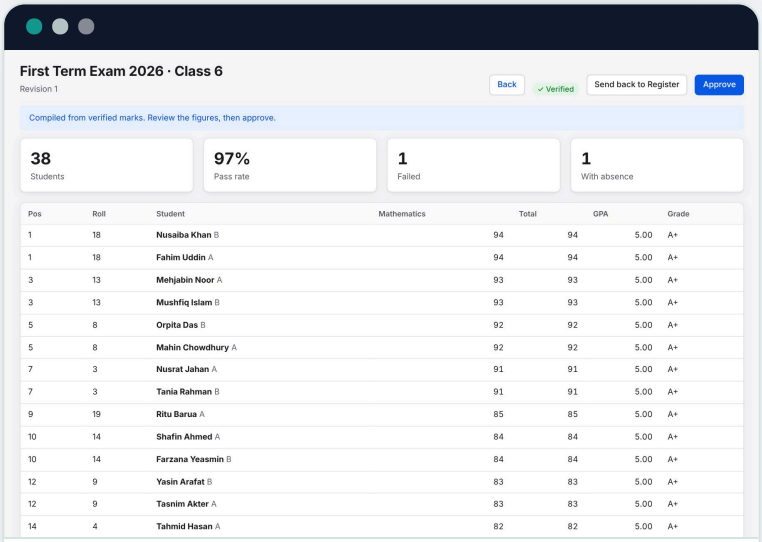
15:31 · STUDENT

Sees the result

A notification arrives; only their own result opens.



Teacher: roll call with present, absent and late.



Headmaster: approval screen with the full result before publishing.

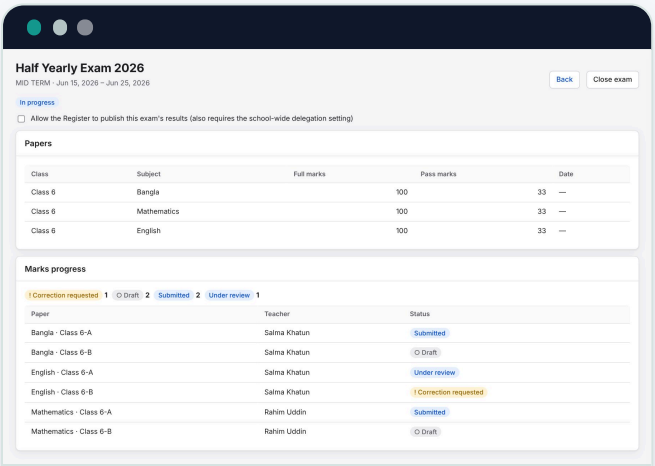
Result: the work that used to take days of sheets and walking between rooms becomes a short chain of clear steps — and each step is visible to the next person.

THE MOST IMPORTANT PROCESS

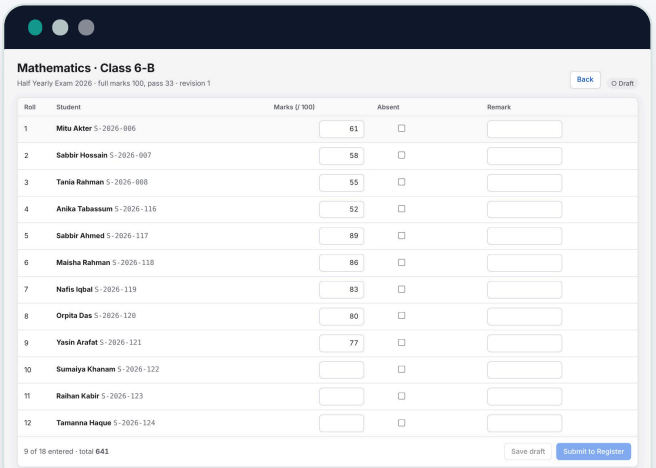
From marks to published result

Results are the most sensitive thing a school publishes. ShikkhaOS makes the path explicit, so every step has an owner and a record.

1. **Register creates the exam**
   Adds papers for each class and subject.
2. **Headmaster authorizes**
   Marks entry opens for the assigned teachers.
3. **Teacher enters marks**
   Invalid marks are rejected on the spot.
4. **Teacher submits**
   The sheet is locked — no silent edits.
5. **Register reviews**
   Can request a correction with a reason.
6. **Register verifies**
   Then compiles the class result.
7. **Headmaster approves**
   Password confirmed again. Can send back.
8. **Result is published**
   Only after approval. Cannot be overwritten.
9. **Students are notified**
   Each sees only their own result.



Exam progress board: every paper and section with its current status.



Teacher marks entry: keyboard-friendly grid, running total, save draft / submit.

Corrections are visible. The Register flags rows and writes a reason; the teacher sees it and resubmits as a new revision. Old versions are kept.

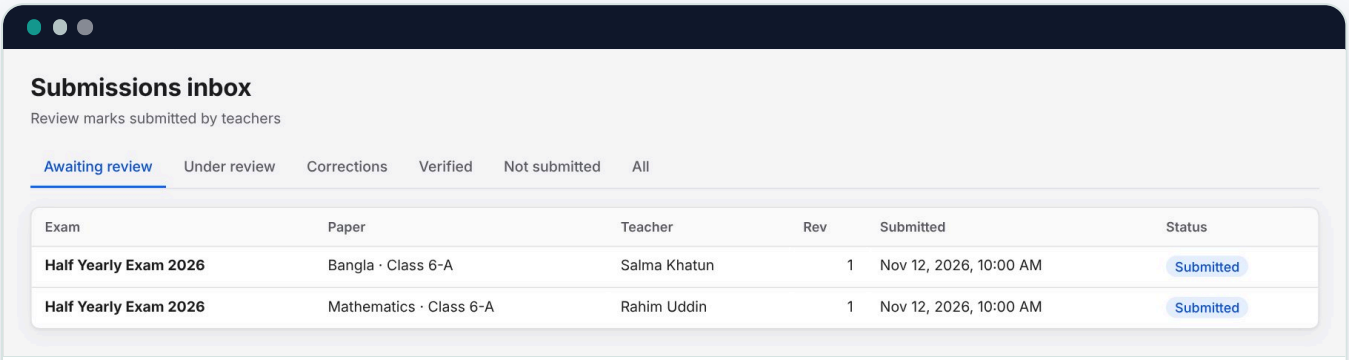
Published means final. A published result cannot be edited. A mistake found later needs a request approved by the Headmaster and creates a new revision — the old one stays on record.

Class tests are separate. Teachers can share class-test marks with their class, but these never change the official result.

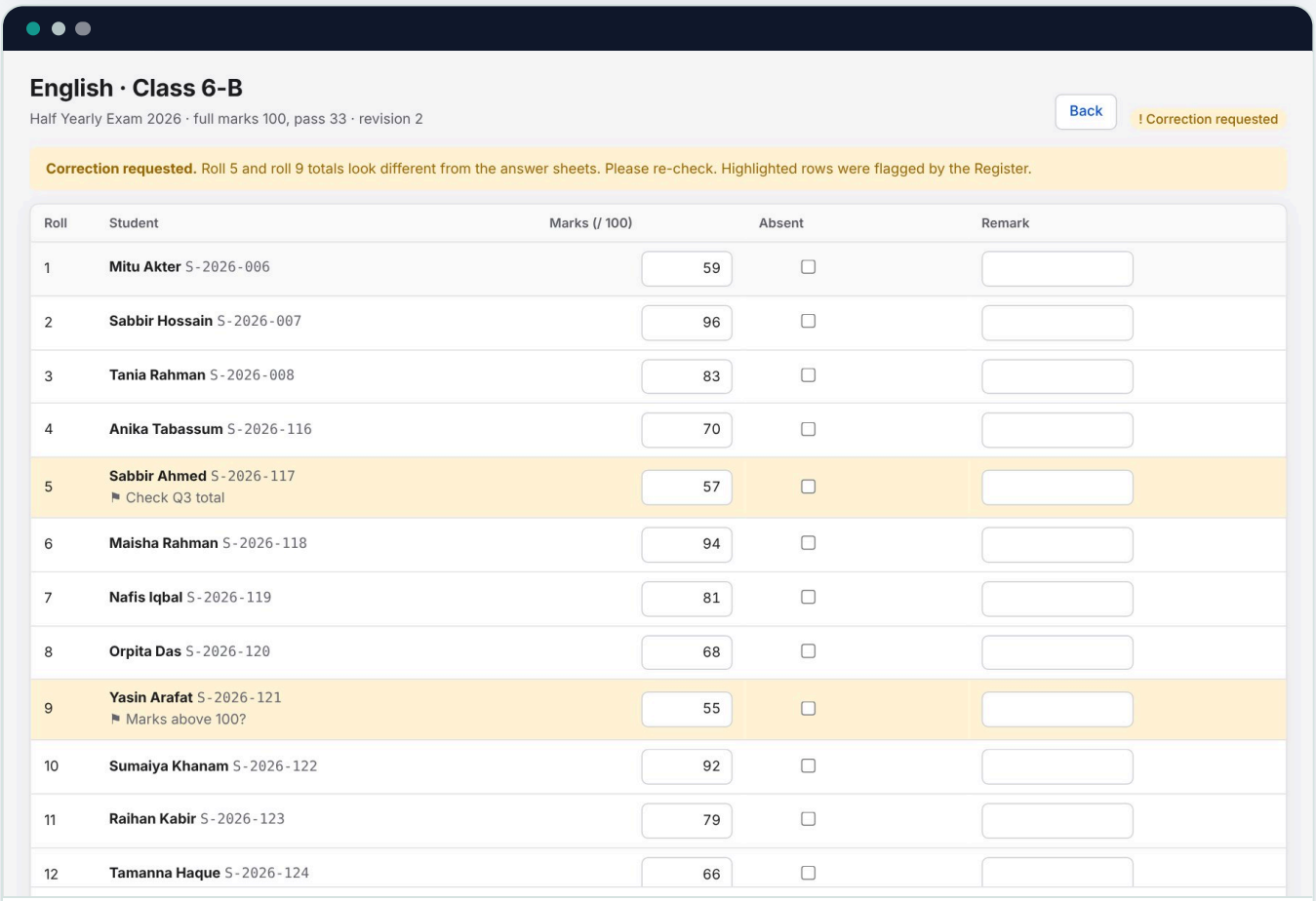
IN USE — REVIEW

The Register checks every submission

Submitted marks arrive in the Register's inbox. Nothing is lost between the teacher and the Register, and nothing can be changed quietly.



Register's inbox: submissions waiting for review, filtered by status (awaiting review, under review, corrections, verified).



Teacher's view after a correction request: the Register's reason and the flagged rows are shown; the teacher fixes and resubmits as a new revision.

Clear statuses
Draft, Submitted, Under review, Correction requested, Verified — always visible.

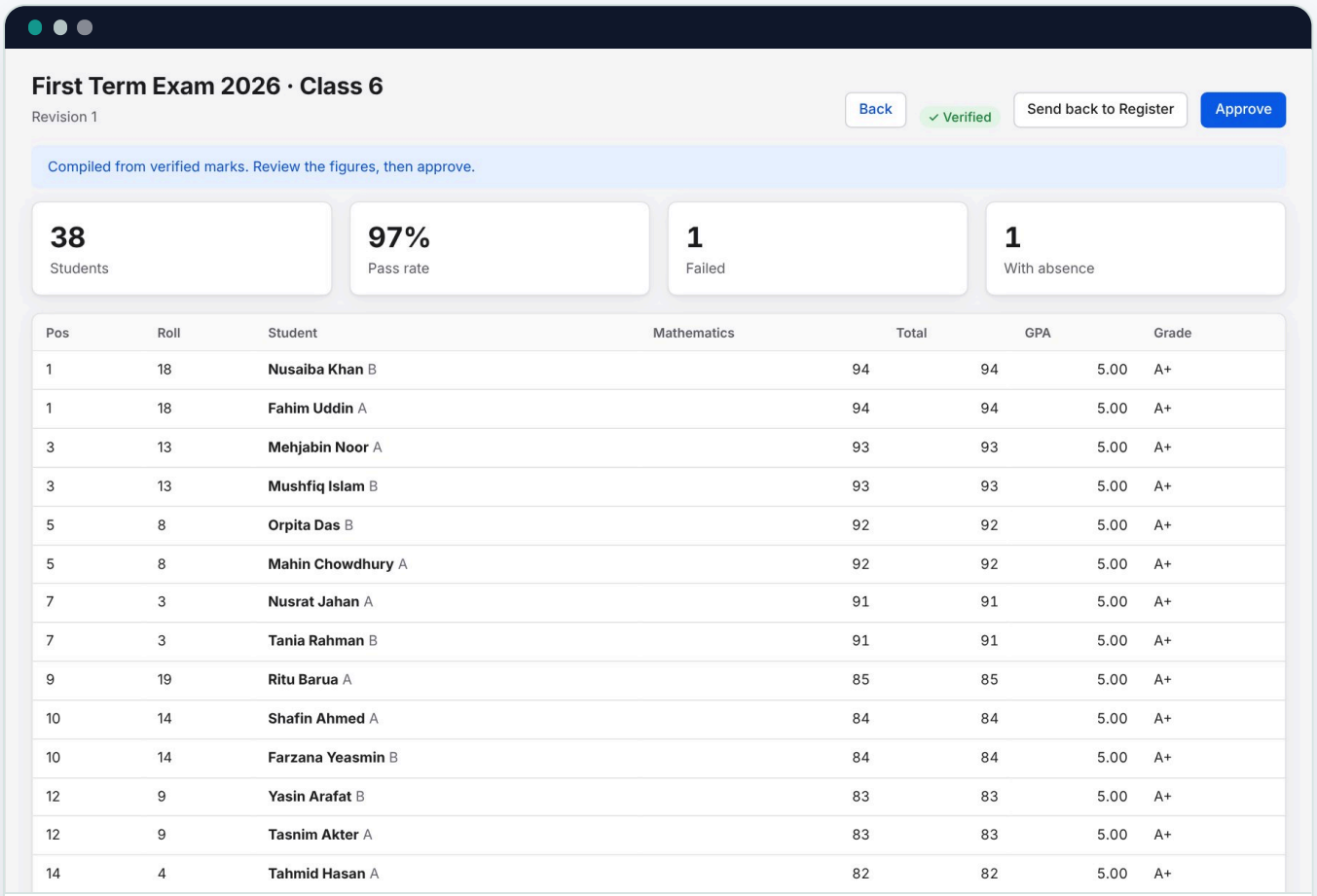
Reasons are required
A correction request must say why; the teacher sees it next to the marks.

History is kept
Each submission keeps every revision and who did what.

IN USE — APPROVAL

The Headmaster decides

Once every paper of a class is verified, the Register compiles the class result. The Headmaster sees the numbers and approves — or sends it back.



Pos	Roll	Student	Mathematics	Total	GPA	Grade
1	18	**Nusaiba Khan** B	94	94	5.00	A+
1	18	**Fahim Uddin** A	94	94	5.00	A+
3	13	**Mehjabin Noor** A	93	93	5.00	A+
3	13	**Mushfiq Islam** B	93	93	5.00	A+
5	8	**Orpita Das** B	92	92	5.00	A+
5	8	**Mahin Chowdhury** A	92	92	5.00	A+
7	3	**Nusrat Jahan** A	91	91	5.00	A+
7	3	**Tania Rahman** B	91	91	5.00	A+
9	19	**Ritu Barua** A	85	85	5.00	A+
10	14	**Shafin Ahmed** A	84	84	5.00	A+
10	14	**Farzana Yeasmin** B	84	84	5.00	A+
12	9	**Yasin Arafat** B	83	83	5.00	A+
12	9	**Tasnim Akter** A	83	83	5.00	A+
14	4	**Tahmid Hasan** A	82	82	5.00	A+

Headmaster's approval screen: pass rate, failures and absences at a glance. Actions: Approve, or Send back to Register with a reason.

Fewer surprises
Pass rate, failures, absences and every student's marks, grade and position are shown before approval.

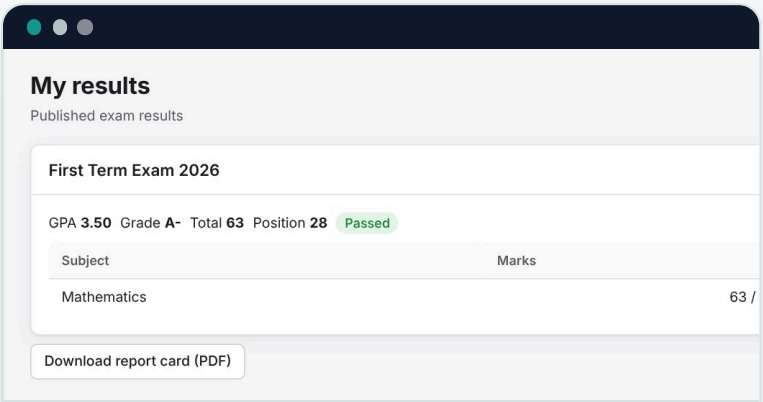
Send back
If something looks wrong, the Headmaster returns the result with a reason. It is re-checked and re-compiled as a new revision.

Two-person control
The person who entered marks cannot verify them; the verifier cannot approve them. Approval needs the password again.

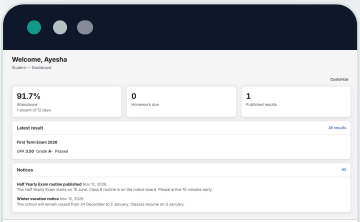
IN USE — STUDENTS

Students see their own results — and only theirs

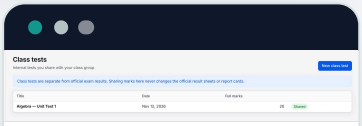
When the Headmaster publishes, each student is notified and can open their result and download a report card.



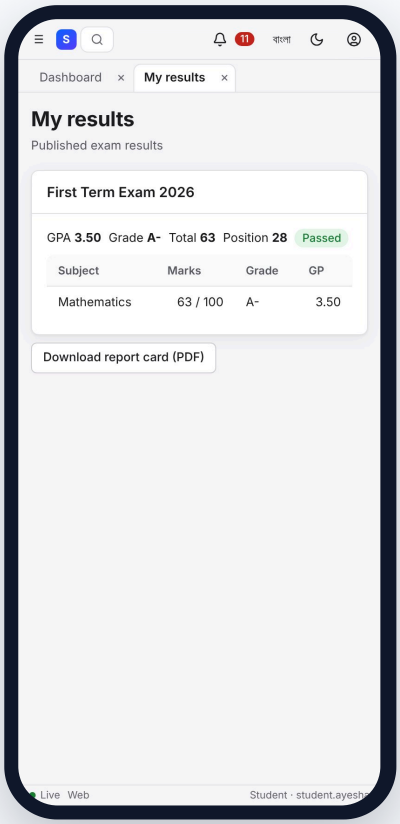
Student's result page (web): subject-wise marks, grade and GPA.



Student dashboard: attendance, homework, latest result, notices.



Class tests: shared with the class, clearly separate from official results.



Result on a phone

Same rules everywhere. A student can open only their own result, attendance and homework — on the phone, the web or any other device.

Note. The phone pictures show the student screens on a phone-sized display. The dedicated Android app has its own native screens with the same features; it is built and tested, and its first installable build is produced by the automated release process (see “Where we are today”).

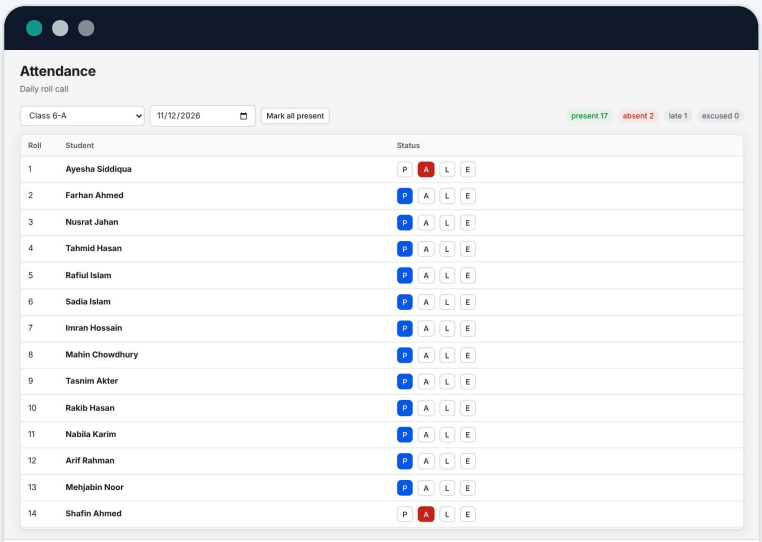
IN USE — DAILY SCHOOL LIFE

Attendance, class tests and homework

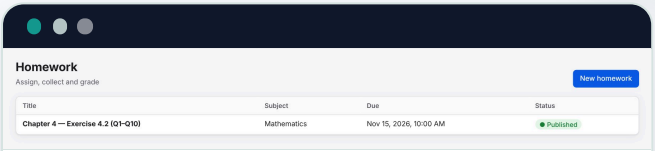
Attendance in seconds

The class teacher marks Present / Absent / Late / Excused for the whole class on one screen. Students marked absent are notified. Past records can be corrected only with a written reason, and the correction is logged.

- ✓ “Mark all present” then change exceptions
- ✓ Attendance % per student and per section
- ✓ Summary report for the Headmaster



Teacher: roll call for Class 6-A.



Teacher: homework with due date; students submit text or photos/files; teacher grades with feedback.

Homework flow

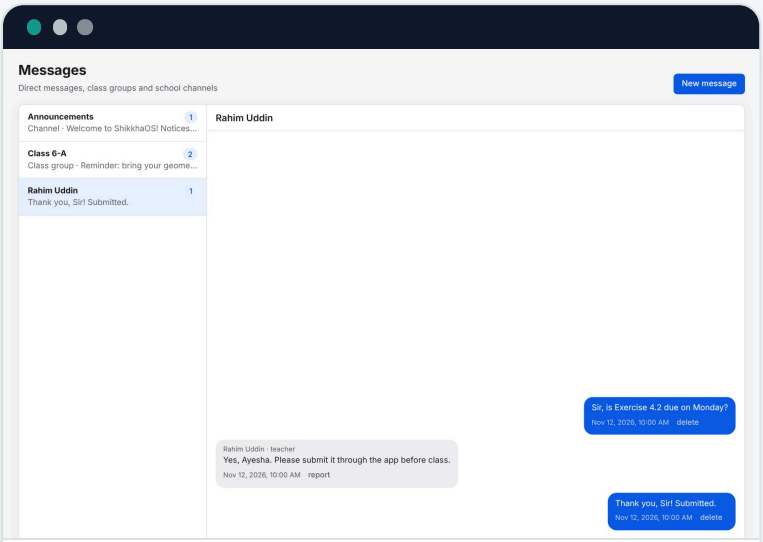
1. Teacher creates and publishes homework for a class.
2. Students are notified, and reminded 24 hours before the deadline.
3. Students submit answers (text, photo or file).
4. Teacher grades and gives feedback; the student is notified.

Late submissions are flagged automatically.

COMMUNICATION AND STUDENT SAFETY

Messages the school can stand behind

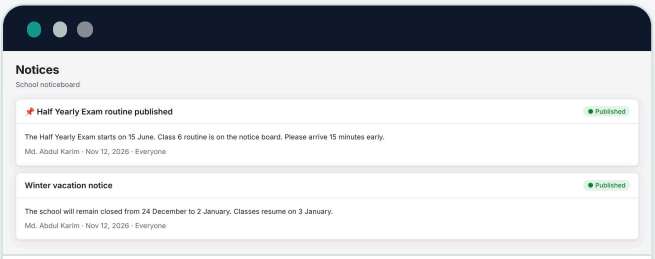
Communication is built into ShikkhaOS so that it follows school rules and protects students.



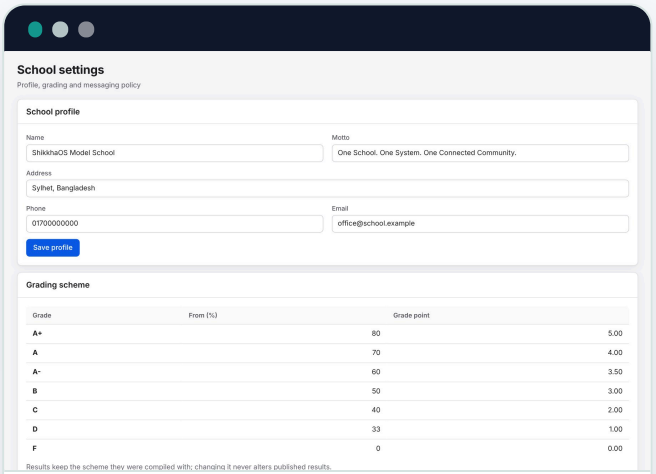
A student asks her teacher a question; the teacher replies. Messages arrive live.

Safety rules built in

- **No private chat between students.** It is switched off and cannot be enabled by mistake.
- Students can message only their **own teachers** and the school office, and the school can switch either off.
- Class groups follow enrollment automatically; students read, and post only if the teacher allows.
- Any message can be **reported**; the Headmaster can hide it or suspend a user's messaging.
- When a moderator reads a student conversation, **that read is recorded**.



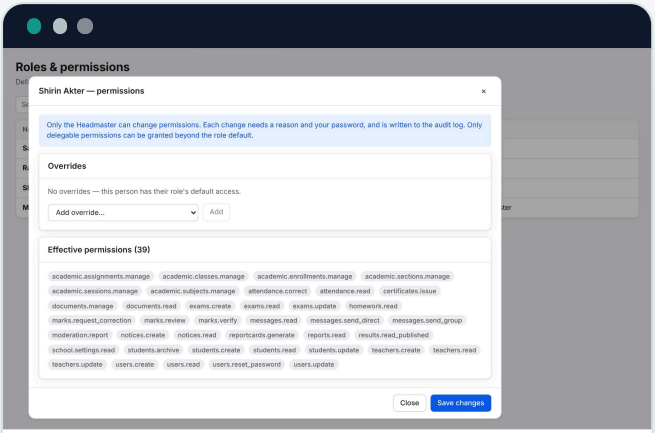
Notices: targeted to everyone, a role, a class or a section; pinned notices stay on top.



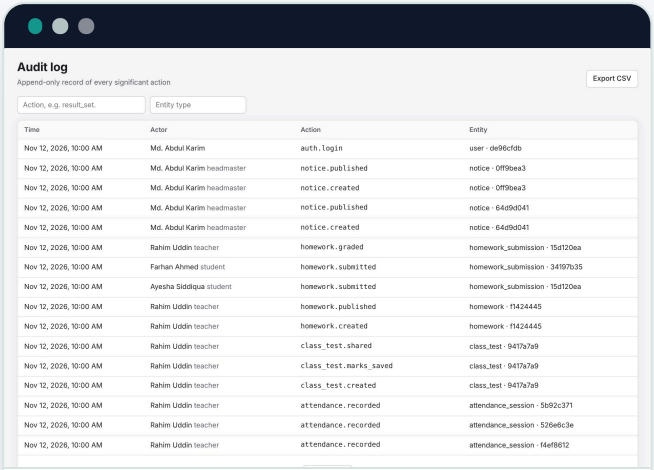
School settings: the Headmaster controls messaging and publishing policy.

CONTROL, TRUST AND RECORDS

Who can do what — and a record of everything



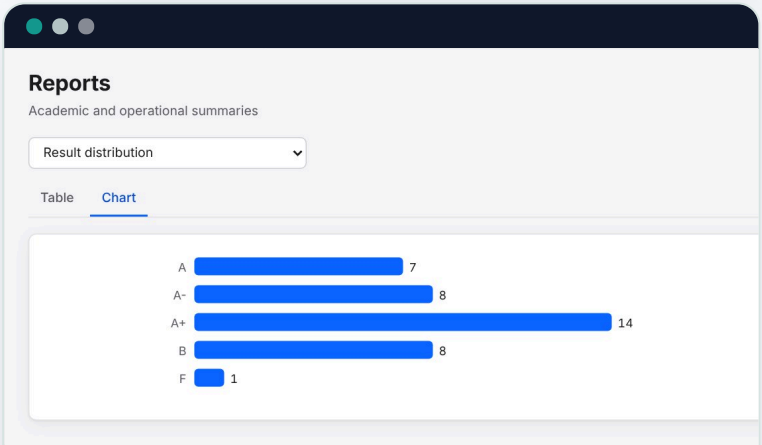
Headmaster: per-person permissions. Every change needs a reason and the password.



Audit log: a permanent, read-only history of important actions.

Protection at a glance

- Passwords stored securely; account locks after repeated wrong attempts.
- Staff and students see only what their role allows.
- Approving or publishing needs the password typed again.
- Published results and the audit log cannot be altered.
- Uploaded files are checked before they are accepted.
- Private information is never put in notification text.



Reports with charts (and a table alternative): grade distribution, attendance, enrollment, exam analysis.

TRUST AND SECURITY

How student data and results are protected

Protection is enforced by the school server — the screens only reflect it. This table lists what could go wrong and what stops it.

What could go wrong	What prevents it
Someone guesses or steals a password	Passwords are stored in a one-way secure form; repeated wrong attempts lock the account; sessions expire and can be ended per device.
A staff member views or edits what they should not	Every request is checked against the person's role and assignment on the server. A teacher cannot open another teacher's class.
A mark is quietly changed after submission	Submitted marks are locked. Corrections require a reason and create a new revision; the audit log records who and when.
One person verifies and approves their own work	Separation of duties is enforced: the verifier cannot approve. Approving and publishing need the password again.
A published result is altered later	Published results are protected at the database level; changes appear only as a new, visible revision.
Students contact each other privately or are contacted by strangers	Student-to-student chat does not exist. Messages are limited to their own teachers and the office; any message can be reported and hidden.
An unsafe file is uploaded	Files are checked for type and size before being accepted.
Private data leaks through notifications	Notification text never contains marks or private details.

Checked, not just promised

The server's security rules are covered by automated tests, a dependency vulnerability audit passes, and the screens were checked against accessibility standards (WCAG 2.1 A/AA).

Still the school’s decision

Where the data is hosted, who holds admin passwords, and how often backups are checked are policy choices for the committee — see the decisions page.

WHERE WE ARE TODAY

What is built, tested — and what remains

Built and tested

- Complete school server and all four user roles.
- Exam → marks → review → approval → publish workflow, including corrections, send-back and revisions.
- Attendance, class tests, homework, notices, messaging with safety rules, reports, certificates and report cards (PDF).
- Web app, Windows desktop app and Android student app (source complete).
- Bulk student import from a spreadsheet (CSV).
- English and Bengali (core screens on web/desktop).

Quality checks run so far

- 48 automated tests on the server (including full result workflow and security checks).
- 30 tests on the apps, plus 8 end-to-end tests that click through the real screens.
- Accessibility check (WCAG 2.1 A/AA) on about 20 screens: no violations found.
- A full exam cycle was run end to end through the real screens.

Still to do before go-live

Item	Why it is open
Windows installer and Android app files	Produced by the automated release process on GitHub; their first run and signing keys are still to be set up and tested.
Phone push notifications	In-app and live notifications work; delivery to a closed phone needs a Firebase setup.
Hosting, backups and monitoring	Depends on the committee's hosting choice (next pages).
Android app in Bengali	Web/desktop are bilingual; the Android app is English-only for now.
Bengali text on printed PDFs	Report-card PDFs currently support English text only; Bengali names need a font upgrade.
Real-school pilot	The system has been tested with demonstration data, not yet with our real classes.

ROLLOUT

A safe, step-by-step start

We recommend starting small, learning, and expanding. Durations are indicative estimates to be confirmed with the committee.

PHASE 1 · ABOUT 2 WEEKS

Prepare

- Choose hosting; set up the school server and backups
- Create Headmaster, Register and teacher accounts
- Import students from a spreadsheet

PHASE 2 · ABOUT 4 WEEKS

Pilot: one class

- Attendance, homework, notices, messages
- Short training for the pilot teachers
- Weekly feedback meeting

PHASE 3 · ONE EXAM CYCLE

First exam on the system

- Marks entry, review and approval for the pilot class
- Run alongside the usual paper process once
- Compare results; fix issues

PHASE 4 · AFTER REVIEW

Whole school

- All classes and teachers
- Student Android app rollout
- Ongoing support and updates

What running it needs

- A small server (school office computer or rented cloud server), always on, with internet.
- A web address (domain) and a security certificate for the site.
- Daily automatic backups, stored away from the server.
- One person per week for the first months for support questions.

An itemized cost estimate will be prepared once the hosting option is chosen.

Risks and how we handle them

Staff unfamiliar	Short hands-on training, pilot first.
Internet outage	Paper fallback during pilot; apps recover automatically.
Data loss	Automatic backups and tested restore.
Wrong access	Roles reviewed by the Headmaster; audit log.

FOR THE COMMITTEE

Decisions we request

Six decisions allow the pilot to begin.

- [ ] **Approve the pilot** in one class (suggested: Class 6, two sections) for one term, as described in Phases 1–3.
- [ ] **Choose where the system runs:** (a) rented cloud server, or (b) a school-owned server. We will prepare costs for the chosen option.
- [ ] **Decide who may publish results.** Default recommendation: **only the Headmaster**. The Register may be allowed per exam if the committee wishes.
- [ ] **Nominate a system owner** (suggested: the Register) and set aside training time for pilot teachers.
- [ ] **Confirm messaging policy:** students may message their own teachers and the school office (recommended), or teachers only.
- [ ] **Confirm data handling:** student records stay on the school's own system; who may export data and how often backups are checked.

What the school provides

- List of students (name, ID, class, section, roll, guardian phone).
- List of teachers and their class/subject assignments.
- Grading rules (current marks-to-grade scale).
- A decision-maker for questions during the pilot.

What the committee receives

- A working system for the pilot, with training.
- Weekly progress notes during the pilot.
- A pilot review: what worked, what to change, whether to proceed.
- Full source and documentation for the school's records.

Committee decision
Date: ___ / ___ / _____

Approved ☐ Approved with changes ☐ Not approved ☐
Chairperson signature: _____

APPENDIX

Everything the system covers

Area	What it includes
Identity and access	Sign-in, forced password change, device list, per-person permissions, password re-confirmation for sensitive actions.
School setup	School profile, grading scale, academic sessions, classes, sections, subjects, teacher assignments, promotions.
People	Student registry (with CSV import and archive), teacher records, enrollments, class groups.
Examinations	Exams and papers, authorization, marks entry, submission, review, correction, verification, compilation, approval, send-back, publishing, revisions.
Daily work	Attendance, class tests (separate from official results), homework and grading.
Communication	Direct messages, class groups, official channels, notices with targeting, notifications, moderation.
Documents	Report cards and certificates (PDF), document library, file attachments with safety checks.
Reports and oversight	Enrollment, attendance, exam analysis, teacher submission status, grade distribution; CSV export; audit log; global search.

Technology, in plain words

The school server is written in Rust, a language known for speed and safety, and stores data in PostgreSQL, a widely trusted database. The apps use common, well-supported tools. Updates are delivered as versioned releases with checksums so the school can verify them.

Document information

Prepared from the working system and its automated test results. Screens show demonstration data. Quality figures are those recorded at the time of writing (November 2026).

Let’s begin with one class.

A small pilot lets teachers, the Register and the Headmaster judge ShikkhaOS on real work — with paper as a safety net — before the whole school commits.

Step 1
Committee approves the pilot

Step 2
We prepare the server, accounts and training

Step 3
One class, one term, weekly review

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